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Part I: Full Speaking Script for Slides 8–16

Paper basis: *Trading Value and Information in MDPs*

Target time: about 20–25 minutes

Let me now move into the mathematical core of the paper. The previous speaker has already introduced the motivation, namely that standard Markov Decision Processes optimize expected reward but ignore the informational cost of specifying actions. My role is to explain how the paper turns that intuition into a formal optimization problem. In this section, I will focus on the main derivation: first the standard value formulation, then the introduction of control information, then the free-energy objective, and finally the modified Bellman equation and the INFO-RL solution. *[Transition from the previous speaker; this corresponds to the beginning of your methodology section, visually introduced around Slides 5–7.]*

First, the authors restate the standard MDP setup. A policy π assigns a probability distribution over actions in each state, and the value of that policy starting from an initial state s_0 is defined as the expected accumulated reward along the trajectory:

$$V_\pi(s_0) = \lim_{T \rightarrow \infty} E \left[\sum_{t=0}^{T-1} R(s_t, a_t) \right].$$

This is familiar, but the reason the paper starts here is very deliberate. The authors want value to serve as one half of a parallel construction. They are about to define another trajectory-level quantity, not for reward, but for the informational cost of control. So before they do that, they first write value in its trajectory form. They also explicitly write the probability of a full trajectory under policy π as the product of policy terms and state-transition terms:

$$\Pr(a_0, s_1, a_1, s_2, \dots \mid s_0) = \prod_{t \geq 0} \pi_{s_t}(a_t) P_{s_t, a_t}(s_{t+1}).$$

This matters because later the information cost will also be expressed over trajectories, and that step depends on this factorization. *[Slide 8: the value function of policy π , and the probability of a whole trajectory under π .]*

The paper then reviews the standard optimality equation. The optimal value function is the unique fixed point of the Bellman optimality principle. In state s , one can write the optimal value as an optimization over the local action distribution $\pi_s(\cdot)$, but because the expression is linear in that distribution, the optimum is always attained at an extreme point of the simplex. That means all the probability mass collapses onto the best action. This is why the optimal standard policy becomes deterministic. In the paper, this is expressed through the usual state-action value function

$$Q^*(s, a) = R(s, a) + \sum_{s'} P_{s,a}(s') V^*(s'),$$

and then the optimal policy chooses the action that maximizes $Q^*(s, a)$. The important conceptual point is that determinism is not being presented as a deep principle of intelligence. It is simply what drops out if we optimize reward alone and assign no cost to the precision of control. *[Slide 9: optimal value function $V^*(s)$, optimal $Q^*(s, a)$, and the deterministic optimal policy.]*

The real turn in the paper comes next, when the authors introduce control information. They imagine that the controller and the actuator are separated by a communication channel. That channel could literally be a radio link to a robot, or more abstractly the pathway from the brain to the muscles. Once we adopt that perspective, precise control is no longer free. To formalize this, the paper defines the one-step control information cost at state s as

$$\Delta I(s) = \sum_a \pi_s(a) \log \frac{\pi_s(a)}{\rho_s(a)},$$

which is the KL divergence between the actual action distribution $\pi_s(\cdot)$ and a default policy $\rho_s(\cdot)$. Usually $\rho_s(a)$ is taken to be uniform over the available actions. The logic here is elegant. If the actual policy is exactly the same as the default policy, then no extra information needs to be transmitted, and the cost is zero. If the actual policy is highly concentrated, then it deviates strongly from the default and carries a higher informational cost. So KL divergence becomes a natural measure of how much extra control information is being used. The paper even gives an intuitive example: with two available actions, a deterministic choice costs one bit relative to a uniform default, whereas a random-walk style policy is essentially free. *[Slide 10: one-step control information cost, assumption of uniform $\rho_s(a)$, and introduction of total control information.]*

The authors then extend this one-step quantity to a full trajectory-level quantity, just as they did earlier for value. Total control information is defined as

$$I_\pi(s_0) = \lim_{T \rightarrow \infty} E \left[\sum_{t=0}^{T-1} \log \frac{\pi_{s_t}(a_t)}{\rho_{s_t}(a_t)} \right].$$

What is especially important is the next rewrite. Because the probability of an action sequence under π factorizes as a product of the $\pi_{s_t}(a_t)$ terms, and similarly under the default policy ρ , the sum of log-ratios over time becomes the log-ratio of entire action-sequence probabilities. So total control information can also be written as

$$I_\pi(s_0) = \lim_{T \rightarrow \infty} E \left[\log \frac{\Pr(a_0, \dots, a_{T-1} \mid s_0; \pi)}{\Pr(a_0, \dots, a_{T-1} \mid s_0; \rho)} \right].$$

This is a very important derivational step, because it shows that the paper is not just adding an arbitrary penalty term. It is defining a trajectory-level divergence in a mathematically consistent way. *[Still Slide 10: the equations for $I_\pi(s_0)$ and the trajectory-probability ratio.]*

Now that both cumulative reward and cumulative information have been defined, the paper combines them into a single objective called free energy:

$$F_\pi(s_0; \beta) = I_\pi(s_0) - \beta V_\pi(s_0).$$

Expanded over the trajectory, this becomes the expected sum of an information term minus a reward term:

$$F_\pi(s_0; \beta) = \lim_{T \rightarrow \infty} E \left[\sum_{t=0}^{T-1} \left(\log \frac{\pi_{s_t}(a_t)}{\rho_{s_t}(a_t)} - \beta R(s_t, a_t) \right) \right].$$

The parameter $\beta > 0$ controls the tradeoff. If β is large, the agent is willing to spend more bits of control to improve reward. If β is small, the agent prefers to stay close to the default policy and save information. The authors also explain that minimizing free energy is equivalent to solving a constrained optimization problem: maximize value subject to a threshold on control information. This is exactly why they connect the framework to rate-distortion theory. It is not a loose metaphor. Structurally, it is the same kind of Lagrangian tradeoff between utility and coding cost. *[Slide 11: free-energy formulation and the constrained optimization interpretation.]*

At this point, the paper has established the objective, but not yet the recursive structure. That is what the next step does. One of the paper's central claims is that free energy, like value, satisfies a Bellman-style optimality equation. The optimal free-energy vector $F^*(s; \beta)$ is defined as the unique fixed point of the operator

$$[BF](s) = \min_{\pi_s(\cdot)} \sum_a \pi_s(a) \left[\log \frac{\pi_s(a)}{\rho_s(a)} - \beta R(s, a) + \sum_{s'} P_{s,a}(s') F(s'; \beta) \right].$$

This equation mirrors standard dynamic programming, but with a crucial difference. In an ordinary Bellman update, we consider immediate reward plus expected future value. Here, the local objective contains three components: the immediate information cost of selecting action a , the negative weighted reward from choosing that action, and the expected continuation free energy of the next

state. In other words, the recursion says that at each state, the optimal local action distribution is the one that balances how costly the action is to specify, how rewarding it is, and what future free-energy consequences it has. *[Slide 13: Bellman's optimality equation.]*

The natural question then is how this minimization is actually solved. This is where the derivation becomes more concrete. The paper proves a lemma that the Bellman operator has the closed form

$$[BF](s) = -\log Z(s; \beta),$$

where the partition function is

$$Z(s; \beta) = \sum_a \rho_s(a) \exp [\beta R(s, a) - E_{s'|s,a} F(s'; \beta)].$$

To understand where this comes from, the key is to look at the local minimization over the distribution $\pi_s(\cdot)$. That minimization has the general structure of a KL-regularized expected cost:

$$\sum_a \pi(a) \log \frac{\pi(a)}{\rho(a)} + \sum_a \pi(a) c(a),$$

subject to $\sum_a \pi(a) = 1$. The standard way to solve this is to introduce a Lagrange multiplier for the normalization constraint, differentiate with respect to $\pi(a)$, and set the derivative to zero. Doing that gives

$$\log \frac{\pi(a)}{\rho(a)} + 1 + c(a) + \lambda = 0,$$

which rearranges to

$$\pi(a) \propto \rho(a) e^{-c(a)}.$$

In the present paper, the local cost term is

$$c(a) = -\beta R(s, a) + E_{s'|s,a} F(s'; \beta).$$

Substituting that in gives

$$\pi_s(a) \propto \rho_s(a) \exp [\beta R(s, a) - E_{s'|s,a} F(s'; \beta)].$$

The normalization constant is exactly $Z(s; \beta)$, and when the optimal $\pi_s(a)$ is substituted back into the objective, the minimized value becomes $-\log Z(s; \beta)$. So this is not just a formal trick. It is the direct solution of the constrained variational problem at each state. *[Slides 13–14: the closed-form solution through the partition function and the Gibbs-like policy.]*

This closed-form result immediately gives the optimal policy:

$$\pi_s(a) = \frac{\rho_s(a)}{Z(s; \beta)} \exp [\beta R(s, a) - E_{s'|s,a} F^*(s'; \beta)].$$

This equation is worth reading carefully, because it contains the logic of the whole framework. The default policy $\rho_s(a)$ acts as the baseline. The term $\beta R(s, a)$ pushes probability mass toward actions with high immediate reward. The continuation term $-E_{s'|s,a} F^*(s'; \beta)$ favors actions that lead to states with low future free energy. The resulting policy is therefore stochastic, but not arbitrarily stochastic. Its softness is exactly calibrated by the tradeoff between reward and information. This is why the paper claims that it gives a principled justification for soft policies. They are not injected after optimization; they are what optimization itself returns once control has a cost. *[Slide 14: closed-form policy and iterative update.]*

The final step is to turn this recursion into an algorithm. Since F^* is the unique fixed point of the Bellman operator, the paper proposes to compute it by iteration:

$$F_{k+1}(s; \beta) = [BF_k](s).$$

Starting from an initial vector, such as all zeros, the algorithm repeatedly updates the free-energy estimates until convergence. Then the optimal policy is recovered from the closed-form expression we just saw. The authors call this procedure INFO-RL. One useful way to explain it orally is to compare it to value iteration. Standard value iteration repeatedly applies a hard-max Bellman backup to $V(s)$. INFO-RL repeatedly applies an information-regularized, soft Bellman backup to $F(s; \beta)$. So conceptually, INFO-RL is not replacing dynamic programming. It is extending dynamic programming to the case where control precision itself is expensive. *[Slides 15–16: INFO-RL algorithm and its interpretation.]*

That concludes the mathematical core. To summarize the logic in one sentence: the paper starts from a standard MDP, defines control information as a KL divergence from a default policy, combines reward and information into a free-energy objective, proves that this objective obeys a Bellman recursion, solves the local optimization in closed form, and obtains a convergent algorithm whose solutions are naturally stochastic. So the central contribution is not just that reward and information are traded off. It is that this tradeoff can be written, solved, and interpreted with the same recursive rigor as an ordinary MDP. *[This is your concluding sentence for the methodology block before handing off to results or to the next conceptual section.]*

Part II: Full Speaking Script for Slides 32–38

Paper basis: *Human Decision Making Balances Reward Maximization and Policy Compression*

Target time: about 15–20 minutes

Now I want to move from the formal theory to the empirical results. What I want to emphasize in this part is not just the authors’ final claims, but how those claims are actually supported by the figures. In other words, for each task, I will first explain what is being manipulated, then what the figure shows, and then why that specific visual pattern supports the policy compression account rather than a standard reinforcement-learning account. The central theoretical prediction is that under a capacity constraint, people should not only become less state-specific. More importantly, they should become biased toward actions with higher marginal probability across states. The experimental results are designed to test exactly that prediction. [*Transition into Slides 32–38.*]

Let me start with Task 1. The question here is whether changing *state frequency* can indirectly bias *action choice*. In the task design, shown on Slide 32, Q1 is the control condition, and Q2 is the experimental condition in which one state, labeled as S^* , appears three times more often than the others. The reward matrix itself still assigns one best action to each state, but because one state occurs much more often in Q2, the action associated with that state becomes more frequent overall. So the key idea is that the manipulation is not directly on the action side. It begins on the state side, but it should propagate into the marginal action distribution $P(a)$. [*Slide 32, panel A: Task 1 setup.*]

Now on Slide 33, panel B, the first thing to notice is the data row. In the reward-complexity scatter on the far left, the blue Q2 points tend to extend slightly upward relative to the red Q1 points, indicating that the Q2 condition reaches somewhat higher average reward. The bar charts make this easier to see. In the average reward bar chart, the Q2 bar is visibly higher than the Q1 bar and is marked with a significance star, which means the increase is statistically reliable. At the same time, the stochasticity bar for Q2 is lower than for Q1, again with a star, and the response-time bar is also lower for Q2, also marked significant. So the empirical pattern in Task 1 is: under the manipulated state-frequency condition, participants earned more reward, behaved less stochastically, and responded faster. Policy complexity itself changes only modestly, but the key point is that performance shifts in a structured direction rather than simply becoming noisier. [*Slide 33, panel B, first row labeled Data.*]

What matters even more is panel C on the right side of Slide 33, because this is where we see the actual action bias. In the Q1 data column, the proportions of the two suboptimal actions within each state are fairly similar. But in the Q2 data column, that symmetry breaks. In state S_2 , the bar for A_1 is higher than the bar for A_3 , and in state S_3 , the bar for A_1 is also higher than the bar for A_2 . Both of those differences are marked with stars. This matters because A_1 is the action whose marginal probability has been increased indirectly by making

its associated state occur more often. So what the figure is showing is not just that participants made more mistakes in Q2, but that their mistakes were systematically biased toward the globally more frequent action. That is exactly what the policy compression model predicts. A standard RL account would mainly predict noise around suboptimal actions, but it would not naturally predict this directional asymmetry between suboptimal choices. [Slide 33, panel C, top row labeled Data.]

The lower two rows of Slide 33 are also important because they compare model predictions. The middle row, labeled Policy compression, reproduces the same qualitative pattern we see in the human data: in Q2, one suboptimal action becomes more likely than the other within a state, and that difference tracks the change in marginal action probability. By contrast, the bottom row, labeled Standard RL & RLWM, gives much flatter and more symmetric predictions across suboptimal actions. So the figure is doing two things at once. It shows the human effect, and it shows that the effect is captured better by the policy compression model than by the baseline alternatives. [Slide 33, panel C, middle and bottom rows.]

Task 2 makes the manipulation stronger and more direct. Instead of changing how often a state appears, the authors now change how often an action becomes optimal across states. On Slide 34, Q1 is still the control condition, but in Q2 some states have multiple optimal responses. The crucial consequence is that one action, A_1 , appears in more optimal mappings across states than the others. That means the marginal action distribution $P(a)$ is no longer uniform in Q2. So if the theory is correct, Q2 should produce an even clearer action-frequency bias than Task 1, because now the bias is driven directly by the structure of optimal actions rather than indirectly by state occurrence. [Slide 34, panel A: Task 2 setup.]

That is exactly what we see in Slide 35. Again, panel B begins with the overall behavioral summary. In the data row, policy complexity is lower in Q2 than in Q1, and this difference is marked significant. At the same time, average reward is clearly higher in Q2, also with a significance star. Stochasticity is lower in Q2, and response time is also lower. This combination is particularly important. Participants are not becoming more complex in order to earn more reward. In fact, they are earning more reward *with lower policy complexity*. That strongly suggests that they are exploiting the redundant structure of the task to compress the policy more efficiently. In other words, because one action works well across multiple states, participants can rely more on that action and still perform well. [Slide 35, panel B, first row labeled Data.]

Panel C of Slide 35 provides the clearest evidence. In the Q1 data column, the proportions of action choices are still relatively balanced within each state. But in the Q2 data column, the bias toward the globally frequent action becomes much stronger. In state S_2 , the A_1 bar is clearly much higher than the A_2 bar, with a star marking significance. In state S_3 , the A_1 bar is again much higher than the A_3 bar, also with a star. So compared with Task 1, the asymmetry is larger and easier to see. This is exactly what we would expect if marginal action frequency is directly shaping behavior. People are not simply confused.

They are preferentially leaning toward the action that is globally more common in the compressed policy. [Slide 35, panel C, top row labeled Data.]

And again, the model comparison strengthens the interpretation. The policy compression row reproduces that strong Q2 asymmetry, whereas the Standard RL & RLWM row stays much more symmetric. So the figure supports two levels of conclusion. First, the human data show a clear action-frequency bias. Second, that bias is quantitatively and qualitatively more consistent with policy compression than with the comparison models. [Slide 35, panel C, middle and bottom rows.]

Task 3 then asks whether time pressure produces the same kind of compression effect. On Slide 36, the reward structure is held fixed, but the response window changes: Q1 gives participants the standard 2-second response period, and Q2 reduces that to 1 second. The theoretical idea is that if policy compression reflects a capacity limit in decision making, then restricting available decision time should force subjects to rely even more on a simplified, lower-complexity policy. [Slide 36, panel A: Task 3 setup.]

Slide 37 shows that this is exactly what happens. In panel B, the first thing to notice is that policy complexity drops sharply from Q1 to Q2, and this difference is significant. Average reward also decreases in Q2, while stochasticity increases and response time becomes shorter, which is of course expected because the experimental condition imposes a shorter window. The important point is that the shorter response period does not simply create random deterioration. Instead, it specifically pushes behavior toward a lower-complexity regime. [Slide 37, panel B, first row labeled Data.]

The strongest evidence again comes from panel C. In the data row, states S_1 and S_2 do not show much change in the relative proportions of the two suboptimal actions. Their Q2-minus-Q1 differences are close to zero. But for state S_3 , the blue difference bar in the Q2-minus-Q1 plot is clearly positive and marked with a star. Looking back at the adjacent proportion plot, this corresponds to an increase in choosing A_1 relative to the alternative under time pressure. This is exactly the state in which the task structure makes the high-marginal action especially relevant. So the result is not a uniform increase in errors across all states. It is a selective increase in the bias toward the globally frequent action in the state where the compression model predicts it should appear. That is a much stronger result than simply saying “time pressure hurts performance.” It shows that time pressure changes the *direction* of errors in a theoretically meaningful way. [Slide 37, panel C, top row Data and the Q2-Q1 difference plot.]

The model comparison tells the same story once again. The policy compression row predicts a positive difference for S_3 , closely matching the human pattern, whereas the Standard RL & RLWM row predicts almost no directional change. So the figure suggests that time pressure does not just add noise. It pushes people to rely more on a compressed policy representation, which in turn increases reliance on the high-marginal action. [Slide 37, panel C, middle and bottom rows.]

The last result asks whether this bias scales with memory load. On Slide 38,

panel A gives an example from the $n_S = 4$ condition, where one action, labeled A_1 , appears more often across state-action mappings than another comparison action, A_2 . The bottom row of the slide then shows the empirical results across set sizes from $n_S = 2$ to $n_S = 6$. In each mini-bar chart in panel B, the bar for A_1 is higher than the bar for A_2 , and in every set size condition there is a significance star. That means the preference for the high-marginal action is robust across different levels of memory load. *[Slide 38, panel B.]*

Panel C summarizes this pattern by plotting $P(A_1) - P(A_2)$ against set size. The curve stays positive throughout and is generally larger at higher set sizes than at the lowest ones. So the main point is not necessarily a perfectly monotonic increase at every single step, but rather that the action bias remains reliably above zero and tends to strengthen as the number of states that must be managed increases. This is exactly what the policy compression account predicts. As the task becomes more memory-demanding, participants rely more on compressed policies, and that compression shows up behaviorally as a stronger preference for the globally frequent action. *[Slide 38, panel C.]*

So if I put these figures together, the empirical logic becomes very clear. Task 1 shows that changing state frequency indirectly produces action bias. Task 2 shows that directly changing action frequency produces an even stronger bias, while at the same time allowing subjects to achieve higher reward with lower policy complexity. Task 3 shows that time pressure pushes subjects toward lower-complexity policies and increases the theoretically predicted directional bias in action choice. Finally, the memory-load analysis shows that this same bias persists and generally strengthens as the state space becomes harder to represent. So the authors' conclusion is not just that humans are bounded. It is that the structure of their errors matches the structure predicted by policy compression. *[Closing synthesis for Slides 32–38.]*

If I wanted to summarize this entire results section in one sentence, I would say that the data support a very specific resource-rational account: when policy complexity is limited, people do not merely become worse at the task; they become biased toward globally frequent actions in a way that is visible in the figures and better explained by policy compression than by the comparison models.